

Purpose: Generate a high current pulse for a short duration to test energy meters under surge/fault conditions.

Applicable meter types:

- Single Phase
- Three Phase Whole Current
- LTCT (Low Tension Current Transformer)

System architecture: Hardware unit (current generator) controlled via software over a serial port connection.

Software requirements:

- Meter-type selector (radio buttons) — Single Phase / Three Phase / LTCT
- **Fire** button → triggers the test
- **View** button → shows the resulting waveform

Test parameters (per meter type):

Meter Type	Current	Tolerance	Duration
Single Phase	1200 A	-3%	10 ms
Three Phase Whole Current	3000 A	-3%	10 ms
LTCT	300 A	-3%	0.5 s

Deliverable: A professional project (with documentation, system description, test methodology) — sounds like this could be a software/GUI + serial communication project, possibly for a university course or PEL-related work.

Complete Project Summary — STOC Test System (Proteus Simulation)

Project Objective

PEL internship project: **Short Time Over Current (STOC)** test system simulation for energy meter testing — applicable to Single Phase, Three Phase Whole Current, and LTCT meters. System uses ATMEGA2560 microcontroller with SCR/thyristor control, software GUI control via serial communication, LCD status display, and waveform capture/display.

Test Parameters (target, for real hardware phase):

Meter Type	Current	Tolerance	Duration
Single Phase	1200 A	-3%	10 ms
Three Phase Whole Current	3000 A	-3%	10 ms
LTCT	300 A	-3%	0.5 s

Components Used

Component	Purpose
ATMEGA2560	Main microcontroller
CRYSTAL (16MHz)	Clock source
CAP (22pF ×2)	Crystal load caps
CAP-ELEC (1μF, C3)	AC coupling capacitor
RESISTOR (various)	Pull-ups, dividers, protection
BUTTON ×4	Phase select (×3) + Fire (×1)
THYRISTOR/SCR	High-current switching element
OPTOCOUPLER-NPN	Isolated SCR gate drive
TRAN-2P2S	Step-down transformer
ALTERNATOR	AC source (simulates mains)
AC Ammeter	Current measurement
LED-RED, LED-YELLOW, LED-GREEN	Phase-selection status indicators
LM016L (LCD 16x2)	Status display
POT (RV1)	LCD contrast control
COMPIM	Serial link to external GUI

Full Pin Connection List (ATMEGA2560)

1. Power & Clock

- VCC pins → +5V
- GND pins → GND
- **AVCC (pin 100)** → +5V (*fixed — was undriven initially*)
- **AREF (pin 98)** → +5V
- XTAL1 → Crystal pin 1 + 22pF cap → GND
- XTAL2 → Crystal pin 2 + 22pF cap → GND
- RESET → 10kΩ resistor → +5V (pull-up), **and separately** RESET → Button → GND

2. Phase Selection & Fire Buttons

- PA0 → Button → common GND rail — Single Phase select
- PA1 → Button → common GND rail — Three Phase select
- PA2 → Button → common GND rail — LTCT select
- PA3 → Button → common GND rail — Fire trigger

3. Status LEDs

- PG0 → 220Ω Resistor → **RED LED** Anode (Single Phase indicator)
- PG1 → 220Ω Resistor → **YELLOW LED** Anode (Three Phase indicator)
- PG2 → 220Ω Resistor → **GREEN LED** Anode (LTCT indicator)
- All 3 LED Cathodes → common GND rail

4. SCR Gate Drive (via Optocoupler-NPN)

- PB0 → 330Ω resistor → Optocoupler **Pin 1** (LED anode)
- Optocoupler **Pin 2** (LED cathode) → GND
- Optocoupler **Pin 6** (Collector) → 1kΩ pull-up → +5V, **and** → SCR Gate (same junction)
- Optocoupler **Pin 4** (Emitter) → SCR Cathode (shared node)
- Optocoupler **Pin 5** (Base) → Not connected

5. Power Stage

- ALTERNATOR → Transformer (TRAN-2P2S) Primary
- Transformer Secondary → SCR Anode
- SCR Cathode → R4 (10Ω, load) → AC Ammeter → back to Transformer Secondary (loop closed)
- SCR Cathode → also wired to Optocoupler Pin 4
- Alternator settings: 12V amplitude, 50Hz (safe demo values; real current scaling done later via transformer ratio)

6. Signal Conditioning for ADC (waveform feedback)

- Load Resistor/Ammeter junction → R5 (100kΩ)
- R5 → R6 (15kΩ) → GND (attenuator divider)
- Same junction → C3 (1μF, negative/hatched side)
- C3 positive side → R8 (10kΩ) → +5V *and* → R9 (10kΩ) → GND (bias divider, ~2.5V reference)
- Same junction (C3/R8/R9) → R7 (1kΩ) → **PC0 (ADC0, pin 53)**

7. Serial Communication (COMPIM)

- **PE1 (TXD0)** → COMPIM RXD
- **PE0 (RXD0)** → COMPIM TXD
- COMPIM has no GND pin (virtual component — normal)
- Settings: Baud rate 9600 (must match firmware), 8 data bits, no parity, 1 stop bit
- RTS/DTR left unconnected (unused handshake pins — expected)

8. LCD (LM016L, 4-bit mode)

- VSS → GND, VDD → +5V
- VEE → POT (RV1) wiper — RV1 terminals to +5V and GND (contrast control)
- RS → PF0, RW → GND, E → PF1
- D4–D7 → PF4–PF7, D0–D3 → not connected
- Backlight A → +5V (via 220Ω), K → GND

Validation Done

- ERC (Electrical Rule Check) run multiple times
 - **Fixed:** AVCC undriven error (connected to +5V)
 - **Remaining (non-critical, expected):**
 - Informational pin-naming warning (PE1/TXD0)
 - COMPIM RTS/DTR undriven (intentionally unused handshake pins)
 - Netlist generated OK, no functional/wiring errors remain
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Current Status: ☐ Schematic Complete & Electrically Verified

Full circuit built: MCU core, phase-select + fire buttons, 3-color LED indicators, isolated SCR gate drive, power stage (Alternator-Transformer-SCR-Load-Ammeter loop), ADC signal conditioning chain, LCD with contrast control, and serial GUI interface — all cross-checked via ERC.

Next Step: Firmware Development

Ready to write ATMEGA2560 firmware (Arduino IDE compatible) covering:

1. Debounced button reading (PA0–PA3)
2. LED indicator logic (PG0–PG2, based on selected phase)
3. Serial command parsing from GUI (phase select + fire command)
4. Timer-based precise pulse generation (10ms for Single/Three Phase, 500ms for LTCT) via PB0
5. ADC reading (PC0) + serial transmission of waveform data to GUI
6. LCD status updates ("Ready", "Firing...", "Done")



